Techfest IIT Bombay 2026–27 | Deliverable 2

Namma Space: Proof of Concept & Web Viewer Link

Official Evaluator Access Document for Interactive 3D Free-Roaming Walkthrough

1. Working Web Viewer Access

Namma Space is a 100% zero-install, browser-based 3D digital twin engine. Evaluators can access and freely roam the reconstructed indoor environments using any modern browser (Chrome, Edge, Firefox, Safari) with WebGL2 / WebGPU support on desktop, laptop, or mobile devices.

PUBLIC LIVE DEMO URL:

`https://namma-space.vercel.app/  (or your live tunnel URL)`

Local Reproduction URL: `http://localhost:5173/` (starts via `npm run dev` inside `frontend/`)

2. Free-Roaming Controls & Capabilities

The viewer loads high-fidelity 3D Gaussian Splats directly with sub-pixel depth precision. Evaluators can navigate seamlessly using standard first-person controls:

- `Click on 3D Canvas` : Locks mouse cursor for full 360° mouse-look
- `W / A / S / D` : First-person forward, left, backward, right walking
- `Shift (Hold)` : Sprint mode (2.5× velocity acceleration)
- `Space / C` : Vertical elevation (Fly up / Fly down)
- 🕹️ `WASD /` 🔄 `Orbit Toggle` : Switches between first-person roam & dollhouse orbit
- `↺ 90° / ↻ 90° / Reset` : In-viewer instant pitch/roll orientation leveling

3. Pre-Loaded Test Scenes for Evaluation

The web viewer features a top navigation bar allowing instant switching across multiple reconstructed rooms:

Scene Name	Capture Source	Gaussian Splats	File Size	Navigation Grid
Ex4 Scene	Living Room & Workspace (Landscape HD)	856,542	27.4 MB	5 cm Obstacle Navmesh
Ex3 Scene	Hallway & Entry Corridor	1,024,878	32.8 MB	5 cm Obstacle Navmesh

Scene Name	Capture Source	Gaussian Splats	File Size	Navigation Grid
Ex2 Scene	Master Bedroom & Balcony	972,683	31.1 MB	5 cm Obstacle Navmesh
Ex1 Scene	Studio Room & Workstation	1,083,150	34.6 MB	5 cm Obstacle Navmesh

4. Point of Interest (POI) & Semantic Search

- **Fuzzy Search:** Type keywords like "Workstation", "Entrance", or "Balcony" into the search bar. Real-time matching is powered by Fuse.js.
- **Fly-To Interpolation:** Clicking any POI smoothly moves the first-person camera directly to the landmark.

5. Evaluator Compatibility Checklist

- ✓ **Browser:** Google Chrome ≥ 113, Microsoft Edge ≥ 113, Mozilla Firefox ≥ 115, or Safari ≥ 17.
- ✓ **Hardware:** Any modern laptop or desktop with integrated or discrete GPU (no RTX card needed on client side).
- ✓ **Connection:** Standard broadband; scene downloads in < 3 seconds on a 100 Mbps connection.